

MIZAN COMBAT ENGINE

GDD v4.0 - Result Set & Pet Mechanics Edition

1. Overview

The **Mizan Engine** is a probability-based turn-based simulation where every turn generates a **Result Set**. This set combines primary character actions with secondary Pet actions. The core philosophy is "Mizan" (Balance), where Physical and Magical attributes are mirrored, and neutral stats act as multipliers for utility layers like evasion and companions.

2. Symmetrical Stat Mapping

Physical Path	Magical Path	Influence on Logic
Strength (STR)	Intelligence (INT)	Damage Floor & High-Tier Action Weight
Dexterity (DEX)	Wisdom (WIS)	Frequency: Multiplies number of actions per turn.
Vitality (VIT)	Spirit (SPI)	Health (HP) vs Mana (MP) Pool and Resistances.
Speed (SPD) - <i>Neutral</i>		Dodge Chance and Turn Priority.
Insight (INS) - <i>Neutral</i>		Pet Attack Logic: Frequency and Success of Pet aid.

3. Cascading Decision Logic & Result Set

In every turn, the engine builds a **Result Set** by following these checks in sequence:

Step 1: Character Action Choice

- **Weapon Draw:** Roll vs STR. If success, choose weapon.
- **Skill Cast:** If Weapon fails, roll vs WIS. If success, check Mana. If Mana fails, action is **Burned**.
- **Fallback:** If all fail, action is **Punch**.

Step 2: Pet Integration (Insight Check)

Independent of the character's success, the Engine performs a Pet check based on **Insight (INS)**. If successful, the Pet's attack is appended to the Result Set with a frequency determined by the Pet's Tameness and INS.

Step 3: Result Set Compilation

The final output is an array of actions to be visualized by the client.

```
[ 3x Hammer, 2x Wolf ]
```

4. Persistence

- **Battle State (Memory):** Stores volatile data like current weapon depletion, spent mana, and pet status. This state is reset after every battle.
- **Account State (DB):** Permanent player data. Results of the battle (Gold, Exp, Elo) are persisted here via the Server Module. Items "lost" in battle via skills are **NOT** removed from the permanent inventory.

5. Implementation Scenarios

Scenario 1: Beastmaster's Might (Phys + Pet)

Stats: 70 STR, 60 DEX, 80 INS. **Loadout:** Hammer, Wolf Pet.

Logic: Weapon draw succeeds (Hammer). Frequency is 3x. Insight check for Pet succeeds. Pet frequency is 2x.

```
[ 3x Hammer, 2x Wolf ]
```

Scenario 2: The Arcane Burst (Magic Only)

Stats: 90 INT, 80 WIS, 10 INS. **Loadout:** Lightning Skill.

Logic: Weapon draw fails. Skill check succeeds (Lightning). Frequency is 3x. Insight check for Pet **Fails**.

[3x Lightning]

Scenario 3: The Stalled Wizard (Failed Cast + Pet)

Stats: High WIS, 0 Mana, 90 INS. **Loadout:** Fireball, Bear Pet.

Logic: Skill check succeeds (Fireball) but **Mana Validation Fails** (Turn Burned). Insight check for Pet succeeds.

[FAILED_CAST, 2x Bear]